DHCP Server Setup & Management

Internal IT Knowledge Base / Standard Operating Procedure (SOP)

Document Type	Internal IT Knowledge Base / User Guide / SOP
Audience	IT Administrators, IT Support, Help Desk, and Non-IT Staff
Environment	Windows Server
Domain	lab.local
Primary Service	DHCP01 - Dynamic Host Configuration Protocol
Related Services	DC01 - Active Directory Domain Services (AD DS), DNS and WDS01

1. Purpose

This document provides a standardized procedure for installing, configuring, testing, and managing a DHCP Server in a Windows Server environment.

DHCP automatically provides network configuration information to client devices, including:

- IP address
- Subnet mask
- Default gateway
- DNS server
- DNS domain name

This eliminates the need to manually configure network settings on every workstation.

2. What Is DHCP?

DHCP (Dynamic Host Configuration Protocol) is a network service that automatically assigns IP configuration to devices connected to a network.

Without DHCP, every workstation would need to be manually configured.

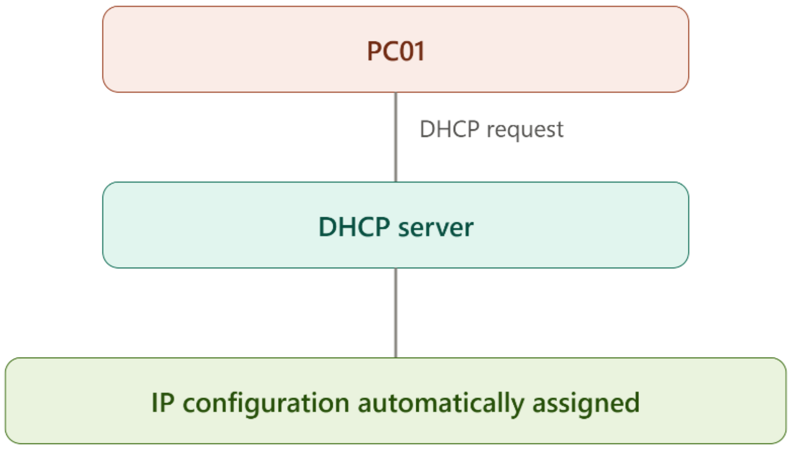
Example

Without DHCP:

```
PC01
IP Address:      192.168.1.110
Subnet Mask:     255.255.255.0
Gateway:         192.168.1.1
DNS:             192.168.1.100
```

The administrator would have to configure these settings manually.

With DHCP:

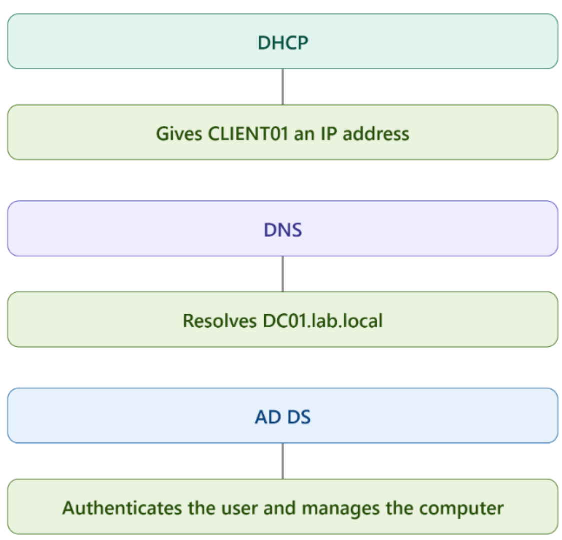


3. DHCP vs DNS vs Active Directory

These services have different responsibilities.

Service	Purpose
DHCP	Automatically assigns network configuration
DNS	Resolves names to IP addresses
AD DS	Manages users, computers, groups, and domains

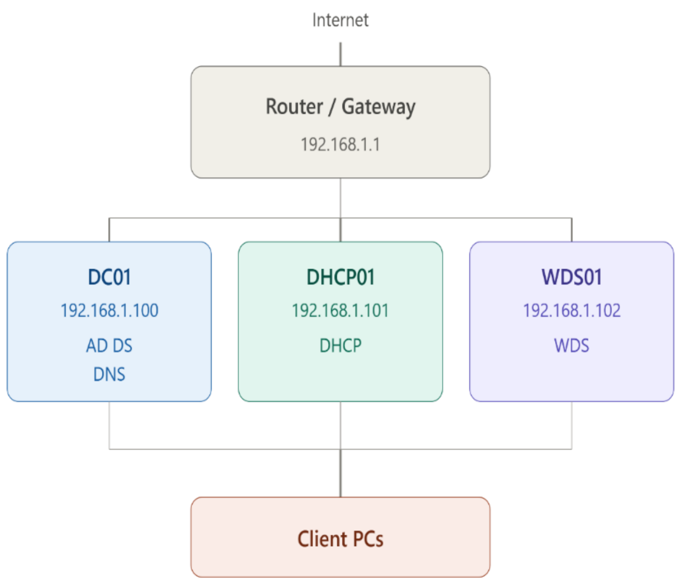
Example:



These services commonly work together in a Windows domain environment.

4. Recommended Lab Architecture

For a small organization or laboratory environment:



Recommended responsibilities

DC01

```
AD DS
DNS
```

DHCP01

```
DHCP
```

WDS01

```
WDS
```

Keeping DHCP and WDS separate is recommended for this lab architecture.

Note: This DHCP document does not cover WDS/PXE deployment configuration.

5. Prerequisites

Before installing DHCP, verify the following.

Server Requirements

- Windows Server installed
- Static IP address configured
- Correct subnet mask
- Correct default gateway
- DNS configured
- Server has a unique hostname
- Server is connected to the correct network
- Administrator privileges are available

Example

```
Server Name:      DHCP01
IP Address:       192.168.1.101
Subnet Mask:      255.255.255.0
Gateway:          192.168.1.1
DNS:              192.168.1.100
Domain:           lab.local
```

6. Configure a Static IP Address

The DHCP server itself should use a static IP address.

Note: Do not configure the DHCP server to obtain its own IP address from the DHCP service it provides.

Go to:

```
Settings
→ Network & Internet
→ Ethernet
```

```
→ IP Assignment
→ Edit
```

Select:

```
Manual
```

Configure:

```
IPv4:         192.168.1.101
Subnet:       255.255.255.0
Gateway:      192.168.1.1
DNS:          192.168.1.100
```

Adjust the values according to the organization's actual network.

7. Install the DHCP Server Role

Step 1 — Open Server Manager

Log in using an account with administrative privileges.

Open:

```
Server Manager
```

Select:

```
Manage
→ Add Roles and Features
```

Step 2 — Installation Type

Select:

```
Role-based or feature-based installation
```

Click:

```
Next
```

Step 3 — Select Server

Select the DHCP server.

Example:

```
DHCP01.lab.local
```

Click:

```
Next
```

Step 4 — Select DHCP Server

Check:

```
DHCP Server
```

When prompted:

```
Add Features
```

Click it.

Continue through the wizard.

Step 5 — Install

Click:

```
Install
```

Wait for the installation to complete.

Note: Do not close Server Manager until the installation has finished.

8. Complete DHCP Post-Installation Configuration

After installation, Server Manager may display a notification.

Click the notification flag.

Select:

```
Complete DHCP configuration
```

Follow the wizard.

If the server is joined to the domain, authorize DHCP using an appropriate domain administrator account.

After successful authorization, DHCP can begin servicing clients.

9. Open the DHCP Management Console

Open:

```
Server Manager
→ Tools
→ DHCP
```

You should see:

```
DHCP
└── DHCP01
    ├── IPv4
    ├── IPv6
    └── ...
```

Expand:

```
DHCP
→ DHCP01
→ IPv4
```

10. Create a DHCP Scope

A scope defines the range of IP addresses DHCP is allowed to distribute.

Right-click:

```
IPv4
```

Select:

```
New Scope
```

The New Scope Wizard will open.

11. Scope Name

Enter a descriptive name.

Example:

```
LAB-NETWORK
```

Description:

```
DHCP scope for company workstations
```

Click:

```
Next
```

12. Configure the IP Address Range

Example network:

```
192.168.1.0/24
```

Configure:

```
Start IP Address:
192.168.1.110

End IP Address:
192.168.1.200
```

Subnet mask:

```
255.255.255.0
```

This allows DHCP to distribute:

```
192.168.1.110
192.168.1.111
192.168.1.112
...
192.168.1.200
```

13. DHCP Exclusions

Exclusions prevent DHCP from assigning specific IP addresses.

For example:

```
192.168.1.100
```

may belong to DC01.

If that address is outside your DHCP range, no exclusion is necessary.

Example:

```
Servers:
192.168.1.100 - 192.168.1.109

DHCP Clients:
192.168.1.110 - 192.168.1.200
```

This is a clean configuration because the DHCP range does not overlap with server addresses.

14. Lease Duration

The lease determines how long a client can use an assigned IP address.

For a normal office network, the default value is generally acceptable.

Example:

```
Lease Duration:
8 days
```

Click:

```
Next
```

15. Configure DHCP Options

Select:

```
Yes, I want to configure these options now
```

Click:

```
Next
```

These options provide additional network configuration to clients.

16. Configure Default Gateway

Select:

```
003 Router
```

Enter the organization's gateway.

Example:

```
192.168.1.1
```

Click:

```
Add
```

Then:

```
Next
```

> **Note:** Verify the gateway address before using it. Do not assume that 192.168.1.1 is always the correct gateway.

17. Configure DNS Server

This is particularly important in an Active Directory environment.

Select:

```
006 DNS Servers
```

Enter the organization's internal DNS server.

For this lab:

```
192.168.1.100
```

Example:

```
DC01
192.168.1.100
```

Click:

```
Add
```

Then:

```
Next
```

Important

Domain clients should normally use the organization's internal AD DNS server, not public DNS servers such as:

```
8.8.8.8
1.1.1.1
```

The internal DNS server is responsible for resolving Active Directory records.

18. Configure DNS Domain Name

Configure:

```
015 DNS Domain Name
```

For the lab:

```
lab.local
```

This allows DHCP clients to receive:

```
DNS Suffix:
lab.local
```

Click:

```
Next
```

19. WINS Configuration

WINS is an older Microsoft name-resolution technology.

For a modern Windows environment, select:

```
Do not configure WINS servers
```

Click:

```
Next
```

20. Activate the Scope

Select:

```
Yes, I want to activate this scope now
```

Click:

```
Next
→ Finish
```

The DHCP scope is now active.

21. Verify the DHCP Scope

In DHCP Manager, expand:

```
DHCP
└── DHCP01
    └── IPv4
        └── LAB-NETWORK
```

You should see:

```
Address Pool
Address Leases
Reservations
Scope Options
Policies
```

22. Verify Scope Options

Open:

```
Scope Options
```

You should see options similar to:

```
003 Router
    192.168.1.1

006 DNS Servers
    192.168.1.100

015 DNS Domain Name
    lab.local
```

The actual values should match your organization's network.

23. DHCP Address Leases

Once clients start requesting addresses, they appear under:

```
Address Leases
```

Example:

```
IP Address       Client
192.168.1.110    CLIENT01
192.168.1.111    CLIENT02
192.168.1.112    CLIENT03
```

This allows IT staff to identify which device currently has a DHCP lease.

24. Test DHCP From a Windows Client

On CLIENT01, open:

```
Command Prompt
```

Run:

```
ipconfig /release
```

Then:

```
ipconfig /renew
```

Check the configuration:

```
ipconfig /all
```

Look for:

```
IPv4 Address
DHCP Server
DNS Servers
DNS Suffix
```

Example:

```
IPv4 Address:     192.168.1.110
DHCP Server:      192.168.1.101
DNS Server:       192.168.1.100
DNS Suffix:       lab.local
```

25. Test Network Connectivity

Test the DHCP server:

```
ping 192.168.1.101
```

Test the domain controller:

```
ping 192.168.1.100
```

Test the gateway:

```
ping 192.168.1.1
```

Test DNS:

```
nslookup lab.local
```

A successful result should identify the internal DNS server.

26. Verify DHCP Server From PowerShell

On DHCP01:

```
Get-DhcpServerv4Scope
```

This displays the configured scopes.

Example:

```
ScopeId        StartRange       EndRange
-------        ----------       --------
192.168.1.0    192.168.1.110    192.168.1.200
```

To view leases:

```
Get-DhcpServerv4Lease -ScopeId 192.168.1.0
```

To view DHCP options:

```
Get-DhcpServerv4OptionValue -ScopeId 192.168.1.0
```

27. DHCP Reservations

A reservation assigns a specific IP address to a specific device based on its MAC address.

Example:

```
Printer01
MAC Address: AA-BB-CC-DD-EE-FF
Reserved IP: 192.168.1.120
```

The device continues to use DHCP but receives the same IP address.

Reservations are useful for:

- Printers
- Network appliances
- Cameras
- Special workstations
- Other devices that need predictable addresses

28. Static IP vs DHCP Reservation

Method	Recommended For
Static IP	Servers and infrastructure
DHCP	Normal workstations
DHCP Reservation	Printers and devices requiring predictable addresses

Example:

```
DC01
Static IP
192.168.1.100

DHCP01
Static IP
192.168.1.101

CLIENT01
DHCP
192.168.1.110

Printer01
DHCP Reservation
192.168.1.120
```

29. Common DHCP Problems

Problem: Client receives 169.254.x.x

This usually indicates that the client did not receive a DHCP lease.

Check:

1. DHCP Server service is running.

2. DHCP scope is active.
3. IP range has available addresses.
4. Client is connected to the correct network.
5. DHCP server is reachable.
6. No firewall or network device is blocking DHCP.
7. There isn't another DHCP server causing conflicts.

Run:

```
ipconfig /release
ipconfig /renew
```

Problem: Client receives an incorrect IP

Check:

```
DHCP → IPv4 → Scope → Address Leases
```

Also check whether another DHCP server is active on the network.

Multiple unauthorized DHCP servers can cause unpredictable results.

Problem: Internet works but domain login fails

Check the client's DNS:

```
ipconfig /all
```

For the AD lab, DNS should point to:

```
192.168.1.100
```

not directly to:

```
8.8.8.8
1.1.1.1
```

Problem: Client cannot find the domain

Run:

```
nslookup lab.local
```

Then:

```
nslookup DC01.lab.local
```

Verify that the responses come from the internal DNS server.

30. DHCP Service Check

On DHCP01:

```
Get-Service DHCPServer
```

The expected status is:

```
Status: Running
```

If necessary:

```
Restart-Service DHCPServer
```

31. DHCP Server Logs

For troubleshooting, check:

```
Event Viewer
→ Applications and Services Logs
→ Microsoft
→ Windows
→ DHCP-Server
```

Also check:

```
Event Viewer
→ Windows Logs
→ System
```

Look for DHCP-related warnings and errors.

32. Security Best Practices

IT administrators should follow these practices:

- Use a static IP for DHCP servers.
- Restrict administrative access.
- Keep Windows Server updated.
- Monitor DHCP leases.
- Avoid unauthorized DHCP servers.
- Document scopes and reservations.
- Use meaningful scope names.
- Keep server IP addresses outside DHCP ranges.
- Back up DHCP configuration.
- Monitor DHCP event logs.
- Do not unnecessarily expose DHCP services to untrusted networks.

33. Recommended IP Address Plan

For the lab environment:

```
Network:
192.168.1.0/24

Gateway:
192.168.1.1

Servers:
192.168.1.100 - 192.168.1.109

DHCP Client Range:
192.168.1.110 - 192.168.1.200
```

Example:

Device	Role	IP Configuration
Router	Gateway	`192.168.1.1`
DC01	AD DS + DNS	`192.168.1.100 Static`
DHCP01	DHCP	`192.168.1.101 Static`
WDS01	WDS	`192.168.1.102 Static`
FILESRV01	File Service	`192.168.1.103 Static`
CLIENT01	Workstation	DHCP
CLIENT02	Workstation	DHCP

34. Important: DHCP and WDS

This document intentionally does not include WDS/PXE deployment configuration.

DHCP's responsibility in this SOP is:

```
IP Address
Subnet Mask
Default Gateway
DNS Server
DNS Domain
```

WDS/PXE deployment should be documented separately as:

```
WDS Server Setup & Windows Deployment SOP
```

That document can cover:

- WDS installation
- WDS configuration
- Boot images
- Install images
- Capture images
- PXE boot
- UEFI configuration
- Client deployment
- Image deployment troubleshooting

This keeps the DHCP and WDS procedures separated and easier to maintain.

35. Final Verification Checklist

DHCP Server

- ☐ DHCP Server role installed
- ☐ DHCP Server has a static IP
- ☐ DHCP Server is authorized
- ☐ DHCP service is running
- ☐ DHCP scope created
- ☐ Scope is active
- ☐ IP range is correct
- ☐ Gateway option configured
- ☐ DNS server option configured
- ☐ DNS domain option configured
- ☐ DHCP leases are being issued

Client

- ☐ Client is configured for automatic IP
- ☐ Client receives an IP address
- ☐ Client receives the correct subnet mask
- ☐ Client receives the correct gateway
- ☐ Client receives the internal DNS server
- ☐ Client receives the correct DNS suffix
- ☐ Client can ping the gateway
- ☐ Client can communicate with the domain controller
- ☐ DNS resolution works

AD Environment

- ☐ DC01 has a static IP
- ☐ DC01 DNS is working
- ☐ DHCP provides DC01 as the DNS server
- ☐ Client can resolve lab.local
- ☐ Client can resolve DC01.lab.local

36. Quick Reference

DHCP Server

```
Server:
DHCP01

IP:
192.168.1.101
```

DHCP Scope

```
Scope:
192.168.1.0/24

Range:
192.168.1.110 - 192.168.1.200
```

DHCP Options

```
003 Router
192.168.1.1

006 DNS Server
192.168.1.100

015 DNS Domain
lab.local
```

Domain Controller

```
DC01
192.168.1.100
lab.local
```

Client

```
IP:
DHCP

DNS:
192.168.1.100
```